

Teacher card

Causes of tectonic hazards / LAUNCH_S2_W01

Outcome: Explain causes of tectonic hazards.

Evidence to retain: F: explanation, map/data interpretation and two exit questions.

Forty minute route

Arrival 4 min; starter 3; model 6; shared task 7; check 2; independent 12; review 3; exit 3.
Pause time is flexible.

Preparation

Choose one independent route. Print the evidence and shared cards, required visual or data sheet, route worksheet and exit. The organiser is one separate A4 page. Keep answers with staff.

Use non-graphic maps and data. A bigger magnitude does not automatically mean greater harm; exposure and vulnerability matter.

Access and regulation

Pre-teach terminology and model evidence use. Chunk sources and data; provide an initial frame, then reduce it. Maintain normal access arrangements. Extend by weighing limitations, alternative explanations and evidence quality. Adults support access and evidence selection without supplying answers.

Offer a predictable start, private response and agreed pause. No cold calling or forced personal disclosure. Rejoin at one clearly named next step. Keep teen-appropriate examples and the same learning goal across response modes.

Use sensitive content warnings where relevant. Do not ask for family details or personal experiences.

Assessment

Use questioning and the exit response to decide the next step. Record the learner's individual contribution and the access support used.

UAS: no approved unit code is supplied. Match the learner's evidence to the centre-selected unit and exact criteria before submission. This pack makes no automatic award claim.

Model and answers

Read these pages before teaching. Hide the model during independent assessment.

Question: What caused the 2011 tsunami?

Evidence: A Japan 11 March 2011: A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Think aloud: Sudden seafloor displacement during the earthquake moved the water above it.

Check: Is magnitude the same as risk? No. Exposure and vulnerability also influence potential harm.

Arrival answers

1. Related to movement of Earth's plates
2. Related to movement of Earth's plates
3. Sudden seafloor displacement during the earthquake moved the water above it.
4. No. Exposure and vulnerability also influence potential harm.

Shared activity reasoning

1. Plate motion and strain — Cause. Motion can accumulate strain along a boundary.
2. Sudden movement and shaking — Event. Energy is released in an earthquake.
3. Warnings and accessible evacuation — Response. These address some consequences and exposure.

Exit responses

What caused the 2011 tsunami? Sudden seafloor displacement during the earthquake moved the water above it.

Is magnitude the same as risk? No. Exposure and vulnerability also influence potential harm.

Teaching and evidence record

Supplied scheme of work teaching sequence: Locate a hazard; build a cause/effects/responses diagram using evidence; interpret one graph/map and make a case-study judgement.

Look for

F: explanation, map/data interpretation and two exit questions.

Accept accurate equivalents, including speech or exact scribing. Check the concept and use of evidence rather than length or handwriting. Record access support separately from content prompting.

BUILD: a concrete example with a reason or demonstrated action. GROW: explain or compare with a relevant detail. LAUNCH: connect evidence and reasoning; use a limitation when the outcome requires it. Keep RE responses non-GCSE.

Media

Original captioned teaching model made for this pack. It demonstrates the supplied reasoning; it is not historical footage, survivor testimony or a documentary. The full transcript and factual source credits are included.

Use manual steps for a calm pace. The silent MP4 has visible captions and a matching text transcript. External archive media is optional and should be previewed before class.

Source credits

USGS 2011 Japan event https://earthquake.usgs.gov/earthquakes/eventpage/official20110311054624120_30/executive

USGS live earthquake feed

https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/2.5_day.geojson

Met Office July 2022 heatwave report https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/weather/learn-about/uk-past-events/interesting/2022/2022_03_july_heatwave.pdf

Source access was checked on 19 September 2026. Restricted links are listed in Sources_and_checks.html; a restriction here is not proof that the page is dead. Offline teaching material remains available.